Mixed Integration Practice

Questions 1–8

Question 1. Find:

a) $\displaystyle\int \frac{1}{\sqrt{1-x^2}}\,dx$

b) $\displaystyle\int \frac{1}{1+x^2}\,dx$

c) $\displaystyle\int \frac{1}{\sqrt{x^2-1}}\,dx$

d) $\displaystyle\int \frac{1}{\sqrt{1+x^2}}\,dx$

e) $\displaystyle\int \frac{1}{1-x^2}\,dx$

Question 2.

a) Show that $\displaystyle\int_{\pi/4}^{\pi/2} 6\cot x\,dx = \ln k$, where k is a positive integer.

b) Show that $\displaystyle\int_{0}^{\pi/2} 6\cot x\,dx$ is undefined.

Question 3.

a) Show that the substitution $u = \tan x$ transforms $\displaystyle\int \frac{1}{25\cos^2 x + \sin^2 x}\,dx$ into $\displaystyle\int \frac{1}{25 + u^2}\,du$.

b) Hence show that $\displaystyle\int_0^{\arctan 5} \frac{1}{25\cos^2 x + \sin^2 x}\,dx = k\pi$, where k is a rational number to be found.

Question 4. Show that $\displaystyle\int_{\sqrt{\ln 2}}^{\sqrt{\ln 5}} \frac{x}{\sinh(x^2)}\,dx = \frac{1}{2}\ln 2$.

Question 5.

a) Find $\displaystyle\int \frac{1}{x^2 - 12x + 61}\, dx$.

b) Hence find the exact value of $\displaystyle\int_5^7 \left(3 - \frac{78}{x^2 - 12x + 61}\right) dx$.

c) The curve $y = 3 - \dfrac{78}{x^2 - 12x + 61}$ intersects the x-axis at the points P and Q.

a) Find the coordinates of P and Q.

b) The curve has a stationary point at the point R. Find the coordinates of R.

c) Find the exact area of the region between the curve, the x-axis, the line $x = 5$ and the line $x = 6$.

Question 6.

a) Find $\displaystyle\int \frac{2x^3 + x^2 + x}{(x+1)(x^2+1)}\, dx$.

b) Hence show that $\displaystyle\int_0^1 \frac{2x^3 + x^2 + x}{(x+1)(x^2+1)}\, dx = 2 - \ln 2 - \frac{\pi}{4}$.

Question 7. By using a suitable substitution, show that:

a) $\displaystyle\int_{-5/2}^{-3/2} \frac{1}{\sqrt{4x^2+20x+34}}\,dx = \frac{1}{2}\ln\left(\frac{2+\sqrt{13}}{3}\right)$

b) $\displaystyle\int_{1/2}^{5/4} \frac{1}{\sqrt{-4x^2+4x+8}}\,dx = \frac{\pi}{12}$

c) $\displaystyle\int_{(\sqrt{3}-3)/2}^{0} \frac{1}{\sqrt{4x^2+12x+18}}\,dx = \frac{\pi}{72}$

d) $\displaystyle\int_{3}^{4} \frac{1}{\sqrt{-4x^2+28x-40}}\,dx = \frac{1}{6}\ln 2$

Question 8. Show that $\displaystyle\int_{-1/\sqrt{3}}^{1/\sqrt{3}} \frac{12x^2}{1-x^4}\,dx = 3\ln(p+\sqrt{q}) + r\pi$, where p, q and r are integers to be found.